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20/4 EATC 7th

Printed Pages – 7 Roll No. .....

328712 (28)

BE (7th Semester)

Examination, April-May, 2013

Branch : Et&T

MICROWAVE COMMUNICATION AND ENGINEERING

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : (i) Part (a) in each question is compulsory.

(ii) Attempt any two parts from (b), (c) and (d).

Unit - I

Q. 1. (a) Define Maximum Usable Frequency. 2

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(b) Explain the mechanism of radiowave bending by the Ionosphere. 7

(c) In the ionospheric propagation, consider that the reflection takes place at a height of 400 km and that the maximum density in ionosphere corresponds to a refractive index of 0.9 at frequency of 10 MHz. Determine the ground range for which this frequency is MUF. Also take earth curvature into account. 7

(d) Explain the duct propagation with the help of neat and clean sketch. 7

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Unit - II

Q. 2. (a) What are the limitations of conventional tubes at microwave frequencies? 2

(b) Describe the Two Cavity Klystron Amplifier with neat diagram. Also explain velocity modulation and bunching process. 7

(c) A reflex Klystron is operated at 9 GHz with a dc beam voltage of 600 V. For 1¾ mode, repeller space length of 1 mm and dc beam current of 10 mA. The beam-coupling coefficient is assumed to be 1. Calculate the repeller voltage, electronic efficiency and output power. 7

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(d) Explain the working principle and construction of magnetron with the help of diagram. 7

Unit - III

Q. 3. (a) What is Transit Time device? Give its types. 2

(b) Explain the principle of operation for Tunnel Diode with the help of Energy-Band diagram. 7

(c) Explain Gunn Diode using Two Valley - Mode Theory. 7

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(d) With suitable diagram, describe the operation of IMPATT Diode. 7

Unit - IV

Q. 4. (a) Define Q of a cavity resonator. 2

(b) Explain Two Hole Directional Coupler. Also establish its S-Matrix. 7

(c) Describe the working principle of circulator and isolator. 7

(d) Briefly discuss the measurement of high and low VSWR. 7

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Unit - V

Q. 5. (a) Draw the typical attenuation response for constant-K, m-derived and composite filters. 2

(b) Explain the process of filter design by the image parameter method. 7

(c) Describe the impedance scaling and frequency scaling for filter transformation. 7

(d) Design a band pass filter having a 0.5 dB equal-ripple response, with N = 3. The

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center frequency is 1 GHz, the bandwidth is

10% and the impedance is 50 Ω. 7

Given data for N = 3,

$g_1 = 1.5963$

$g_2 = 1.0967$

$g_3 = 1.5963$

$g_4 = 1.000$

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